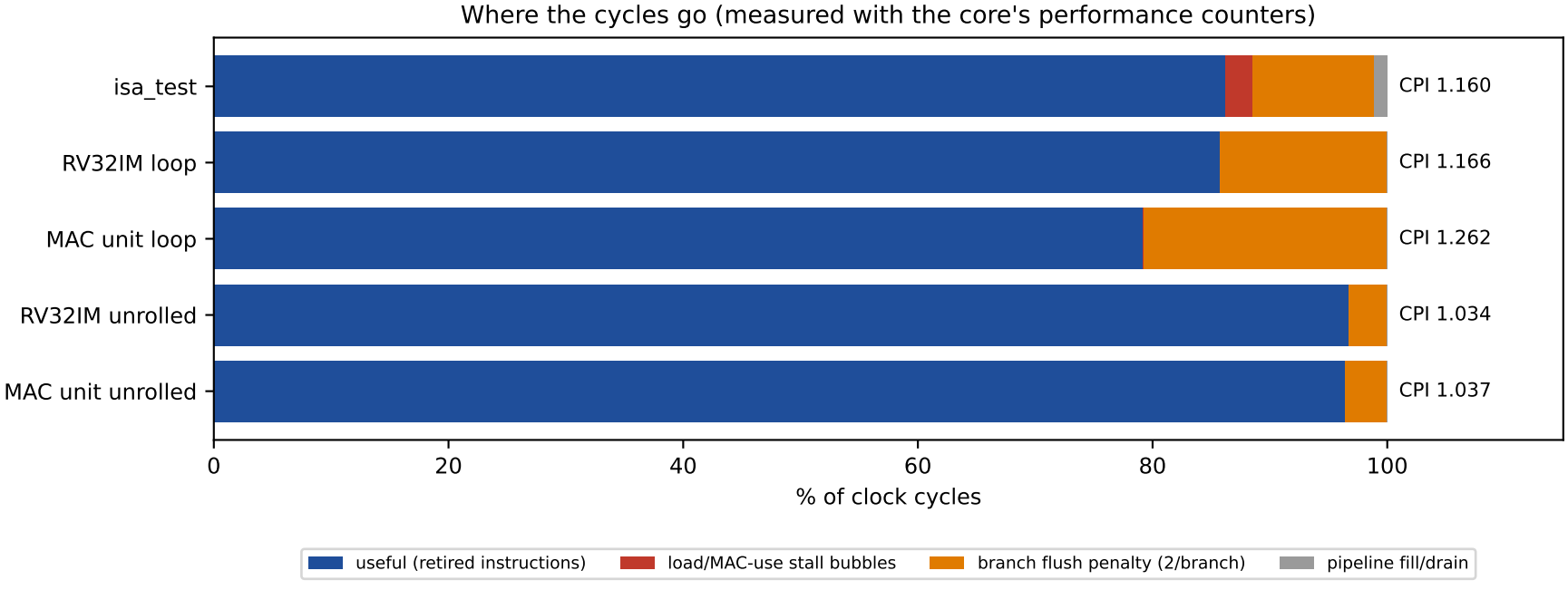
Where the cycles go (measured with the core's performance counters)
isa_test
RV32IM loop
MAC unit loop
RV32IM unrolled
MAC unit unrolled
CPI 1.160
CPI 1.166
CPI 1.262
CPI 1.034
CPI 1.037
0
20
40
60
80
100
% of clock cycles
useful (retired instructions)
load/MAC-use stall bubbles
branch flush penalty (2/branch)
pipeline fill/drain